

# Chapter 6: Cellular Respiration

## Introduction to Energy Production

Cellular respiration is the metabolic pathway that breaks down glucose and produces ATP. The stages of cellular respiration are:

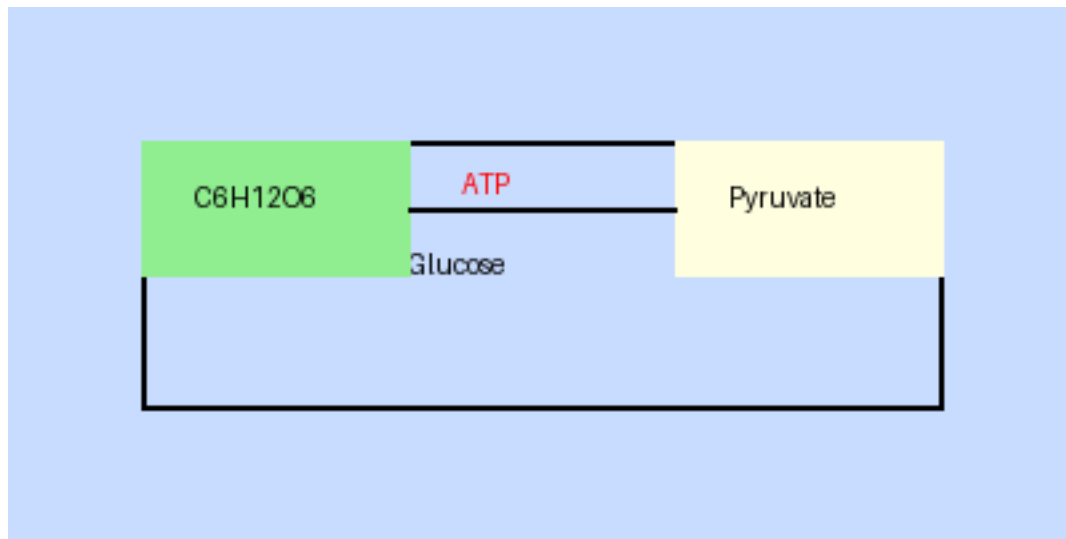
In this chapter, we will explore each stage of cellular respiration in detail. We will learn how cells convert the chemical energy of glucose into ATP.

The process occurs in three main stages:

1. Glycolysis - occurs in the cytoplasm
2. Krebs Cycle - occurs in the mitochondrial matrix
3. Electron Transport Chain - occurs in the inner mitochondrial membrane

Each stage plays a crucial role in extracting energy from glucose molecules.

Figure 6.1: The Glycolysis Pathway



Glycolysis is the first step in cellular respiration. It occurs in the cytoplasm and does not require oxygen.

Key points:

- Glucose (6 carbons) is split into 2 pyruvate molecules (3 carbons each)
- Net gain: 2 ATP molecules
- Produces 2 NADH molecules
- Occurs in the cytoplasm of all living cells

The process can be divided into two phases:

1. Energy Investment Phase: Uses 2 ATP to phosphorylate glucose
2. Energy Payoff Phase: Produces 4 ATP and 2 NADH